# Whole-Home Wi-Fi Expansion: Option Analysis

RT-BE92U network: AiMesh node vs. third-party mesh, UniFi and GL.iNet Flint

Edition v1.7 · 2026-10-08 · Ryan Musante

| | | |
|---|---|---|
| **RT-BE92U**<br>AiMesh node to add | **$249.99**<br>MSRP for one unit | **10G or 2.5G**<br>Ethernet backhaul; wireless as fallback |
| **2,882 Mb/s**<br>Host PHY ceiling at 160 MHz | **7 of 20**<br>Options that pass every must (§7) | **V1–V8**<br>Field checks; worksheet in Appendix A |

Facts about this network are as stated, not re-measured; vendor figures are as listed; prices are listing snapshots (§13).

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## 1 Verdict

**Buy one ASUS RT-BE92U and add it as an AiMesh node on Ethernet backhaul: 10G when the primary's 10G port is free and the run is Cat6A (Cat6 is assured only to 37 m), otherwise 2.5G. Use MoCA over coax when no Ethernet run exists, and wireless MLO backhaul only when neither exists.**

**Confidence.** High that an AiMesh node is the right fix: it follows from the stated configuration and vendor specifications. Medium on throughput: no RT-BE92U pair has been measured, so check V2 (§10) settles it.

### 1.1 Why this option

- Only AiMesh nodes extend coverage inside the primary's own roaming domain while leaving its services untouched: DoT to AdGuard, plaintext DHCP DNS, the USB SSD Samba share and the nvram configuration.
- The node carries the 6 GHz-only SSID the GTR9 Pro joins and needs no controller or PoE.
- The primary's 2×2 5 GHz and 6 GHz radios cap wireless backhaul at 2 spatial streams per band, so a costlier node such as the BT10 or BQ16 Pro adds at most receive diversity at range.
- It costs $249.99 against $279.99 for the ZenWiFi BT8, which Amazon sold at $189.99 on 2026-04-27. Only the RT-BE90U also meets every requirement, $10 less at list but with no 10G port (§4, U15).

### 1.2 Do not buy

- **ZenWiFi BD4/BD5, ZenWiFi BE5000 Pro, RT-BE82U and RT-BE58 Go:** dual-band, with no 6 GHz radio.
- **ZenWiFi BN12:** Wi-Fi 8 at £529.99, a UK pre-order with no ETA; its gains need Wi-Fi 8 clients, and US sale needs FCC conditional approval (U12).
- **Deco, Orbi or eero kits:** they cannot keep the Asuswrt services and one roaming domain without discarding the RT-BE92U's radios (§5.1).
- **Ubiquiti UniFi:** the same 2×2 class on 5 and 6 GHz, plus a controller and PoE for ceiling APs; justified only by a VLAN-driven rebuild (§5.2).
- **GL.iNet Flint 3 and Flint 4:** routers, not nodes; as a second access point they split roaming (§5.3), and the Flint 4 needs FCC conditional approval for US sale (U12).

# 2 Baseline network

| Element | Current state | Consequence for this decision |
|---|---|---|
| Router | ASUS RT-BE92U on stock Asuswrt, by choice; Asuswrt-Merlin supports the unit but is not used | The node must be an ASUS AiMesh unit on stock firmware |
| Router ports | 10G WAN/LAN and 2.5G WAN/LAN, both auto-sensing, plus 3 × 2.5G LAN | Wired backhaul at 10G or 2.5G (§6.1) |
| Router radios | 2.4 GHz 3×3; 5 GHz 2×2; 6 GHz 2×2; channels up to 320 MHz | Wireless backhaul capped at 2 spatial streams per band |
| Link aggregation | Not offered on the firmware reviewed | Above 2.5G needs the 10G port at both ends |
| Upstream DNS | AdGuard ad-block tier as WAN DNS; router DoT on since 2026-07-31; AdGuard validates DNSSEC | Stays on the primary |
| Client DNS | Plaintext DNS from the router via DHCP, by design; the host pins no upstreams, DoT or DNSSEC settings | The node must not touch DHCP or DNS; no host-side DoT |
| Priority client | GTR9 Pro: MediaTek MT7925 (Filogic 360, 2×2 EHT) with a 160 MHz hardware ceiling; joins the 6 GHz-only SSID ASUS_68-6G at 160 MHz EHT | The node needs a 6 GHz radio; 320 MHz and Wi-Fi 8 headroom is unusable here |
| Storage | Lexar ES3 ext4 USB SSD on the router, shared over Samba | Stays on the primary |
| Segmentation | Smart Home Master sets up to three SSIDs; ASUS’s VLAN FAQ omits the RT-BE92U; Merlin adds no VLAN on this class | VLAN needs are the one case for a platform change (§12) |
| Wi-Fi security | WPA/WPA2-Enterprise is unavailable in AiMesh mode | Blocks the plan only if Enterprise authentication is in use (U3) |

# 3 Requirements

Must-level requirements gate the options; should-level requirements rank those that pass.

| ID | Requirement | Level | Rationale |
|---|---|---|---|
| R1 | Extend 6 GHz coverage while carrying the host’s 6 GHz-only SSID | Must | The priority client uses 6 GHz only |
| R2 | Keep the primary’s Asuswrt stack: DoT and AdGuard, DHCP DNS, USB SSD share, nvram configuration | Must | Deliberate design |
| R3 | One roaming domain: a single system steering clients between access points | Must | AiMesh uses ASUS’s own roaming and band steering, described as proprietary rather than 802.11k/v/r-based; separate systems steer independently |
| R4 | Backhaul of 2.5 Gb/s or more: Ethernet link rate or measured wireless throughput | Should | Keeps backhaul at or above realistic host goodput; host PHY ceiling 2,882 Mb/s (§13) |
| R5 | No controller, PoE injectors or extra appliances | Should | Operational simplicity |
| R6 | Low incremental cost | Should | A single-unit purchase is preferred |

# 4 AiMesh node candidates

| Node | 6 GHz radio | Wired backhaul | Notes | Price snapshot (US unless noted) | Verdict |
|---|---|---|---|---|---|
| **RT-BE92U** | Yes, 2×2 | 10G or 2.5G | Identical to the primary: same radios, ports and firmware line | MSRP $249.99 | **Recommended** |
| RT-BE90U | Yes, 2×2; 5,764 Mb/s | 2.5G: one WAN, three LAN | 2×2 on 5 and 6 GHz like the primary; no 10G port; RTINGS: top speeds very similar to the RT-BE92U | List $239.99; Amazon $189.99 (2026-04-27) | Alternate (2.5G) |
| ZenWiFi BT8 | Yes, tri-band | 2.5G: one WAN, one LAN | Upright, 18.6 cm tall; seven internal antennas; MLO backhaul | MSRP $279.99; Amazon low $189.99 (2026-04-27), average $271.54 | Alternate |
| ZenWiFi BT10 | Yes, 4×4; 11,529 Mb/s | 10G: 10G WAN/LAN and 10G LAN | Second 10G port; extra streams unusable against the primary's 2×2 radios | SG: S$1,076 per 2-pack | Overspend |
| ZenWiFi BQ16 Pro | Yes, two 4×4; 11,529 Mb/s each | 10G: 10G WAN/LAN and 10G LAN | Quad-band; the second 6 GHz radio is a dedicated backhaul between BQ16 Pro units; extra streams unusable against the primary's 2×2 radios | Newegg $659.99; ASUS store $899.99 | Overspend |
| ZenWiFi BD4/BD5 | No, dual-band | BD5: 2.5G | Cannot carry the 6 GHz-only SSID; BD5 wireless backhaul ≈265 Mb/s | Not checked | Reject |
| ZenWiFi BE5000 Pro | No, dual-band | 2.5G: two WAN/LAN | Same specification as the ZenWiFi BD5, under a Best Buy name; cannot carry the 6 GHz-only SSID | Best Buy 3-pack only, $299.99 | Reject |
| RT-BE82U | No, dual-band | 2.5G: one WAN, four LAN | 4×4 5 GHz at up to 160 MHz; cannot carry the 6 GHz-only SSID (U16) | MSRP $219.99; B&H $199.00 | Reject |
| RT-BE58 Go | No, dual-band | 2.5G WAN; 1G LAN | Travel router powered over USB-C; RTINGS: poor range; cannot carry the 6 GHz-only SSID | MSRP $159.99; B&H $119.00 | Reject |
| ZenWiFi BN12 | Yes, Wi-Fi 8; 11,529 Mb/s | 10G: 2 × 10G, 3 × 2.5G | Joins AiMesh with most existing ASUS routers; Wi-Fi 8 gains need Wi-Fi 8 clients | UK: £529.99, pre-order, no ETA | Defer |

**Matched pair.** A second RT-BE92U shares the primary's radios, ports and firmware line; the cost is that a model-specific firmware regression reaches both units at once (U5). The RT-BE90U is the nearest unmatched node: the same 2×2 6 GHz class for $10 less at list, capped at 2.5G wired backhaul, with MLO wireless backhaul documented only between RT-BE90U units (U15).

# 5 Alternatives outside AiMesh

## 5.1 Third-party mesh kits

TechRadar still rates the TP-Link Deco BE63 its top mesh system (guide updated 2026-02-17), a sound pick for a from-scratch install. On this network, any third-party kit hits one of two dead ends:

| Integration | Consequence | Fails |
|---|---|---|
| Kit replaces the RT-BE92U | The DoT and AdGuard chain, the nvram configuration and the USB SSD share are rebuilt on the kit's platform or dropped | R2 |
| Kit in AP mode behind the RT-BE92U | Two Wi-Fi systems steering independently, or the RT-BE92U's Wi-Fi 7 radios switched off and replaced with purchased ones | R3 or R6 |

Verdict: reject. Kits were not priced, because the only variant that meets R2 and R3 discards the RT-BE92U's own Wi-Fi 7 radios (R6).

## 5.2 Ubiquiti UniFi

Ubiquiti's consumer AmpliFi Alien is listed as discontinued since 2023-06-15; the Wi-Fi 7 gateways found for this report, the Express 7 and the Dream Router 7, are UniFi devices.

| Device | Radios | Ports and power | Role | Price snapshot |
|---|---|---|---|---|
| UniFi Express 7 | 2×2 on 2.4, 5 and 6 GHz; 5,700 Mb/s at 6 GHz | 10G WAN, one 2.5G LAN; USB-C 5 V/5 A power | Gateway with a controller for 30 devices, wired AP or wireless-backhaul mesh AP | $219.99 and $244.99 at one US retailer |
| UniFi U7 Pro | 2×2 per band; 5,765 Mb/s at 6 GHz | One 2.5G port; PoE+ only, no power input; up to 21 W | Ceiling AP; standalone mode limits features | MSRP $200 |
| UniFi Dream Router 7 | 2×2 on 2.4, 5 and 6 GHz | 10G SFP+ and 2.5G WAN; 3 × 2.5G LAN including one PoE port; 15.4 W budget | Gateway with a controller; VLAN segmentation; microSD for camera and call recordings | Not checked (US) |

| Mode | What it takes | Outcome | Fails |
|---|---|---|---|
| Standalone APs, no controller | U7 Pro plus a PoE+ injector per AP | No 802.11r/k/v roaming between APs | R3, R5 |
| Managed APs behind the RT-BE92U | A UniFi console or the self-hosted UniFi Network Server (Windows, macOS, Linux); PoE+ per AP | Two Wi-Fi systems unless the RT-BE92U's radios are switched off | R3 or R6; R5 |
| Full swap: UX7 or UDR7 plus APs | Rebuild DNS: UniFi Encrypted DNS takes a DNS-over-HTTPS stamp, the way AdGuard's own UniFi guide sets it; DoT is not documented. No USB storage share: RTINGS reports no USB port on the UDR7 (conflict: U8). The UDR7's 15.4 W PoE cannot feed a U7 Pro drawing up to 21 W | One roaming domain plus VLAN segmentation, which ASUS does not list for the RT-BE92U | R2 |

Verdict: lateral on radios (2×2 on 5 and 6 GHz, like the RT-BE92U), heavier to run and destructive to the Asuswrt stack in its only roaming-clean form. Revisit only if VLANs become a requirement that Smart Home Master SSIDs cannot meet (§12).

## 5.3 GL.iNet Flint 3 and Flint 4

GL.iNet's Flint routers run OpenWrt-based firmware on a Qualcomm (Flint 3) or MediaTek (Flint 4) platform. They are routers, not AiMesh nodes: on this network a Flint can only run as a second Wi-Fi system behind the RT-BE92U or replace it.

| Device | Radios | Ports and USB | Firmware and modes | Price snapshot |
|---|---|---|---|---|
| Flint 3 (GL-BE9300) | 2×2 on all bands; 688 / 2,882 / 5,765 Mb/s; MLO; RTINGS: good range, speed suits a 2 Gb/s connection | Five 2.5G ports: WAN, WAN/LAN and three LAN; USB 3.0; no 10G | OpenWrt 23.05 fork, admin panel 4.x, AdGuard Home; Router, Access Point, Extender and WDS modes, with the wired web GUI unreachable in AP or WDS mode | $189.90 direct; $143 to $158 via GL.iNet's AliExpress store |
| Flint 4 (GL-BE14000) | 688 / 4,323 / 8,646 Mb/s; six antennas; MLO behavior unconfirmed | 10G RJ45 and 10G SFP+ WAN/LAN, 2.5G WAN/LAN, 3 × 2.5G LAN, 4 × 1G LAN; USB 3.0 Type-A and Type-C | OpenWrt-based; MediaTek quad-core A73, 2 GB RAM, 64 GB eMMC; VPN up to 1.5 Gb/s | $299 pre-order (2026-08-04), GL.iNet store only; needs US approval (U12) |

**Fit against the Asuswrt stack (R2)**

| Asuswrt service | GL.iNet equivalent | Fit |
|---|---|---|
| Router DoT to AdGuard DNS | Encrypted DNS mode offers DNS over TLS and DNS over HTTPS; a custom server such as a private AdGuard DNS endpoint needs SSH and is not officially supported | Partial |
| USB SSD Samba share | The File Sharing app serves USB storage over Samba (SMB2); both Flints have USB 3.0 | Yes |
| Ad blocking | AdGuard Home runs on the router | Yes |
| Segmentation | An isolated guest network since firmware 4.5; full VLANs on the Flint 3 only through LuCI or UCI scripts, which community authors warn can crash-loop the router | VLANs, no GUI |
| AiMesh roaming domain | “Mesh” appears in the Flint 3 specifications; the roaming method and partner models are undocumented (U13) | Unknown |
| nvram configuration | OpenWrt UCI; nothing carries over | Rebuild |

| Mode | What it takes | Outcome | Fails |
|---|---|---|---|
| Flint 3 as AP or extender behind the RT-BE92U | Cable to a 2.5G LAN port; the Flint 3 has no 10G port | A second Wi-Fi system steering on its own; wired GUI access lost in AP or WDS mode | R3 |
| Flint 3 or 4 replaces the primary (RT-BE92U becomes an AP) | Rebuild DoT (custom server via SSH), Samba, DHCP and firewall; VLANs via UCI | Coverage still comes from a second, independent system, and the Asuswrt stack is gone | R2, R3 |
| Flint pair: router plus extender or mesh | 2 × Flint 3 at $189.90 | One vendor domain; roaming method unverified (U13) | R2; R6 |
| Flint 4 in any role | A model announced after 2026-03-23 | US sale needs FCC conditional approval (U12) | Availability |

Verdict: reject for Wi-Fi expansion. A Flint adds coverage only as a second, independently steering system (R3) or by replacing the primary (R2). Keep the Flint 3 in view as the cheaper VLAN-capable rebuild: $189.90, with DoT, Samba and ad blocking built in and VLANs by command line rather than GUI (§12).

# 6 Backhaul engineering

## 6.1 Paths

Choose the path from what the house already has (Figure 1), then free the port the primary needs for it (Figure 2). Figure 3 sets the resulting ceilings against the only wireless backhaul figures that were measured.

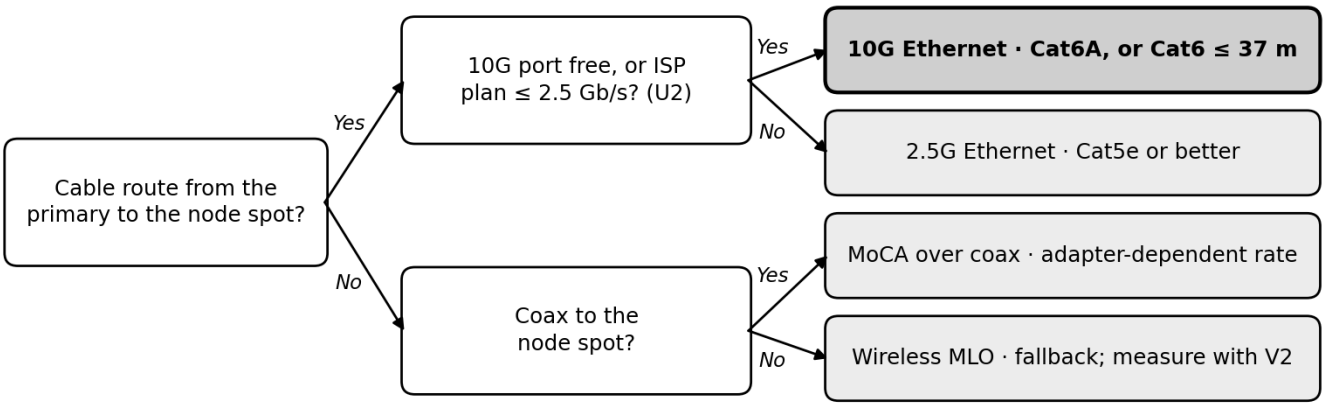



*Figure 1. Backhaul path selection. The 10G branch needs the primary’s 10G port free (Figure 2); Cat6 is assured only to 37 m and reaches 55 m only where alien crosstalk allows.*

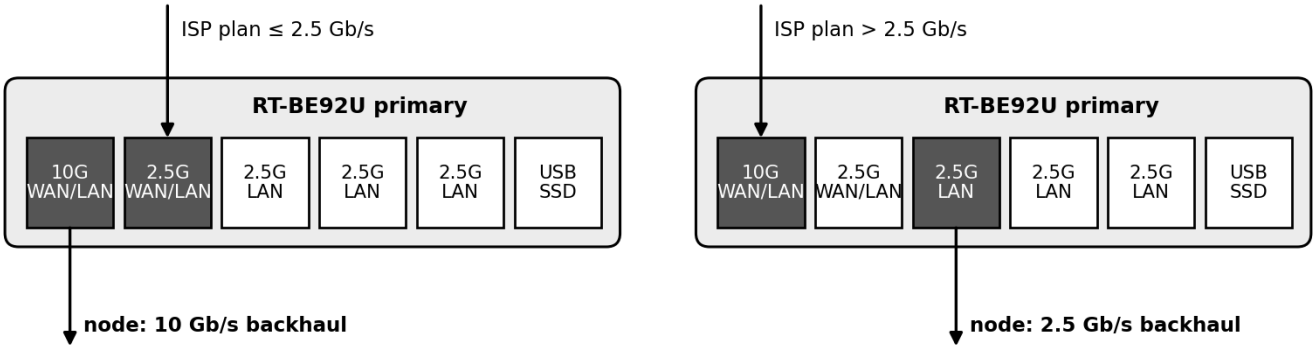



*Figure 2. Port plan on the primary (schematic; physical port order not shown). Left: an ISP plan of 2.5 Gb/s or less moves WAN to the 2.5G WAN/LAN port and frees 10G for the node. Right: a faster plan keeps WAN on 10G, and a 2.5G LAN port feeds the node.*

| Path | Ceiling | Cabling | Notes |
|---|---|---|---|
| 10G Ethernet | 10 Gb/s link rate | Cat6A to 100 m; Cat6 55 m typical, 37 m under worst-case bundling | Needs the primary's 10G port free (Figure 2) |
| 2.5G Ethernet | 2.5 Gb/s link rate | Cat5e or Cat6 to 100 m | Any LAN port; under the host's 2,882 Mb/s PHY ceiling, but real Wi-Fi goodput runs well below PHY |
| MoCA over coax | Adapter-dependent (not assessed) | Existing coax | Wired-like option; one owner runs AiMesh nodes on MoCA behind an RT-BE92U |
| Wireless MLO, 5 and 6 GHz | Measured: BT10 pair ≈727 Mb/s average, ≈1 Gb/s peak; BD5 pair ≈265 Mb/s | None | The node's single 6 GHz radio carries the host link and the backhaul leg (§6.2) |

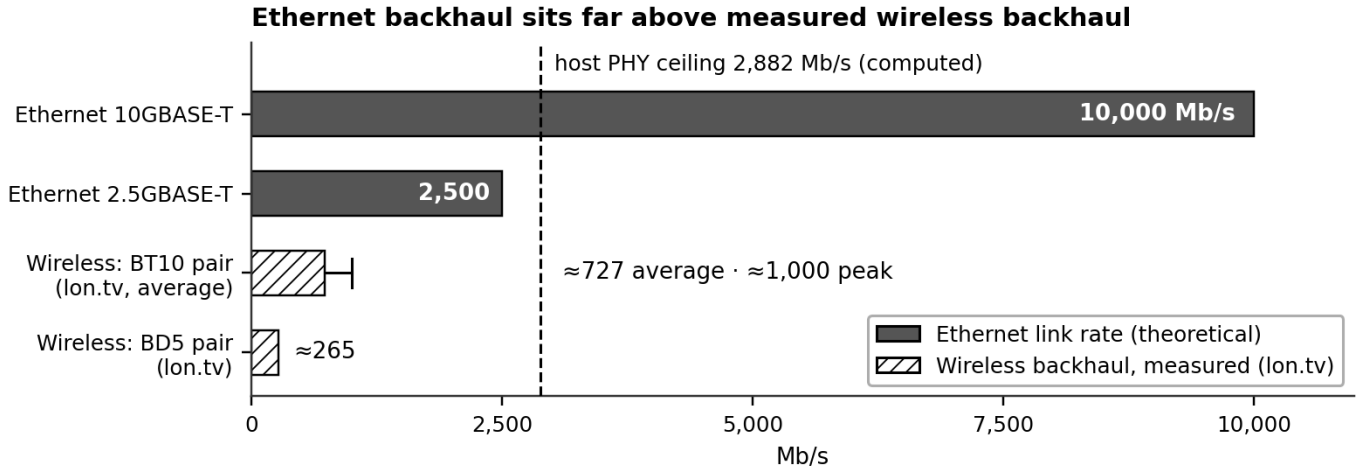



*Figure 3. Backhaul ceilings. Solid: Ethernet link rates (theoretical). Hatched: lon.tv's measured ZenWiFi wireless backhaul; no RT-BE92U pair was measured. Dashed: host PHY ceiling (computed).*

## 6.2 Airtime on tri-band units

Each tri-band unit here has one 6 GHz radio. With wireless backhaul, the node's 6 GHz radio carries both the host's link and the backhaul leg on the same channel, so host traffic crosses that channel twice unless MLO moves the backhaul to 5 GHz. Wired backhaul removes the second crossing (Figure 4). The quad-band BQ16 Pro adds a second 6 GHz radio for backhaul, documented only between BQ16 Pro units (U15).

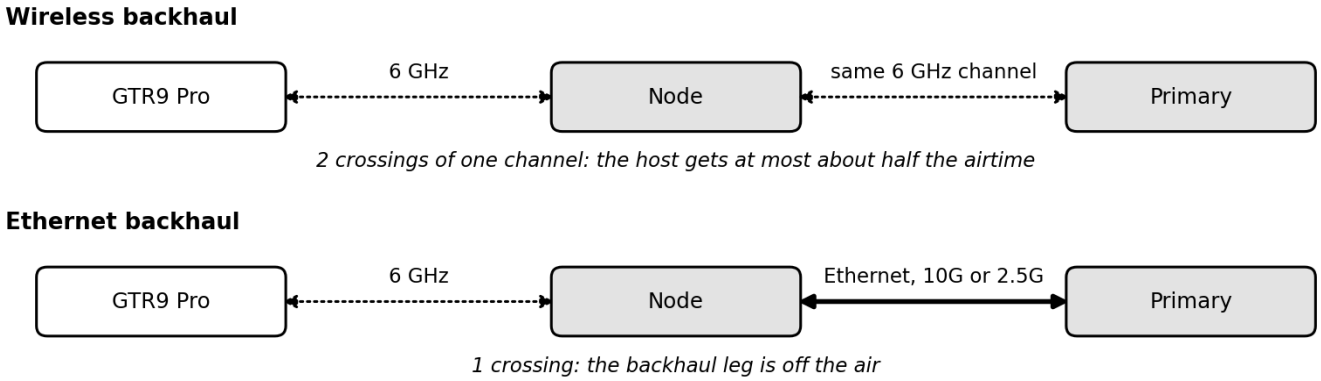



*Figure 4. Airtime on a one-radio node (conceptual). With wireless backhaul each host frame crosses the 6 GHz channel twice, so the host gets at most about half the channel's airtime (inference); Ethernet takes the second hop off the air.*

## 6.3 Headroom against the host ceiling

Figure 5 rates every candidate 6 GHz radio with a vendor rating against the host. Capacity right of the dashed line is unusable by the GTR9 Pro, whose MT7925 tops out at 160 MHz.

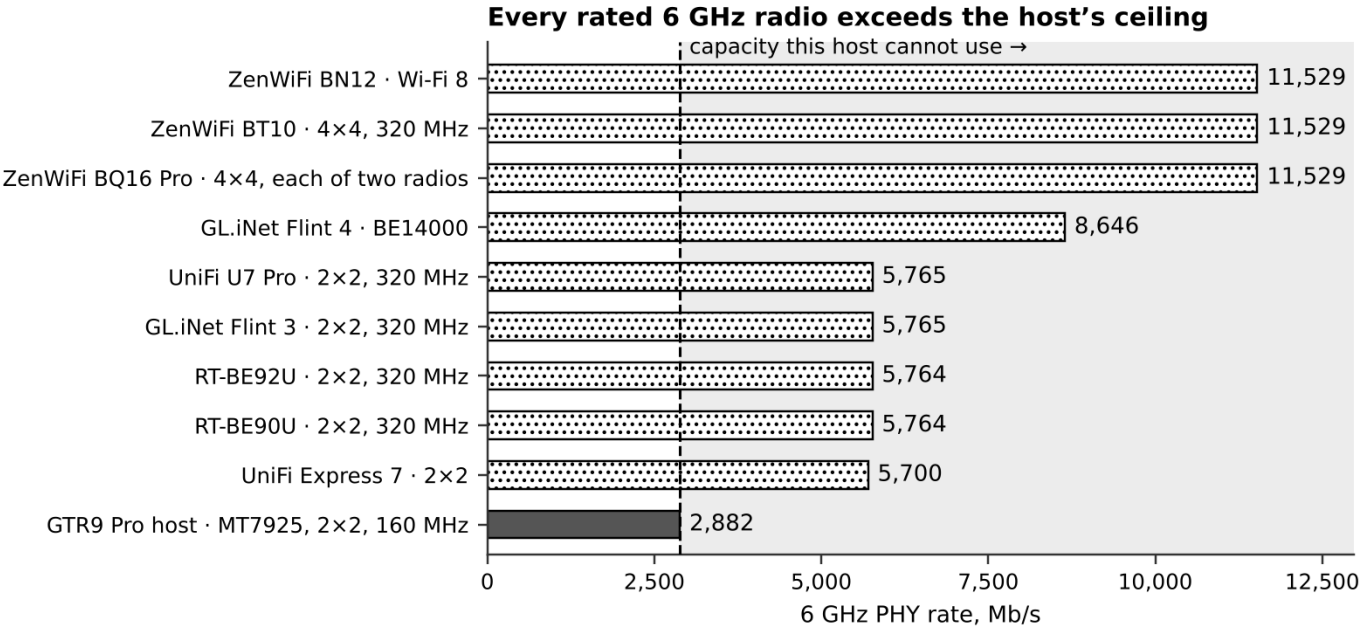



*Figure 5. 6 GHz PHY ceilings, vendor-rated as listed by ASUS, GL.iNet and retailers; host value computed for EHT, 2 spatial streams, 160 MHz, MCS 13, 0.8 µs GI (§13). The BQ16 Pro value is per radio. Not shown: the dual-band nodes, which have no 6 GHz radio, and the BT8 and UniFi Dream Router 7, which have no separate 6 GHz rating.*

# 7 Decision matrix

Scored against §3: PASS, PART and FAIL are judgments from the facts above; n/c means not checked. Figure 6 ranks the options; the matrix that follows shows each judgment, with shading that repeats the label for black-and-white print.

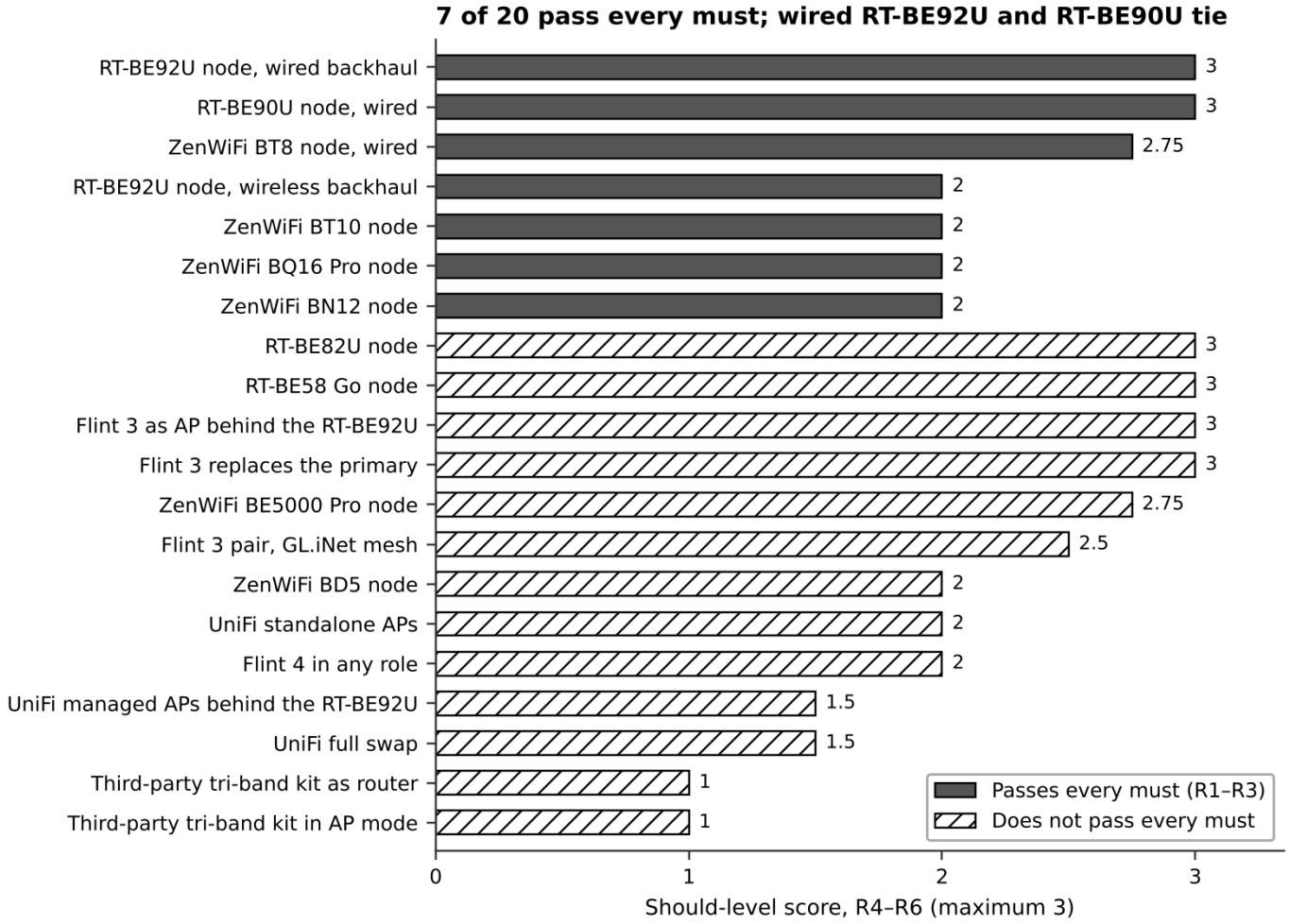



*Figure 6. Should-level score (R4–R6) per option: PASS 1, PART ½, FAIL or n/c 0; cost LOW 1, LOW+ ¾, MED ½, HIGH or n/c 0. Hatched options fail, only partly meet or were not checked against a must-level requirement (R1–R3) and are out regardless of score.*

| Option | R1 6 GHz | R2 stack | R3 roam | R4 ≥2.5G | R5 no extra gear | R6 cost | Verdict |
|---|---|---|---|---|---|---|---|
| RT-BE92U node, wired backhaul | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | LOW | **Recommended** |
| RT-BE92U node, wireless backhaul | **PASS** | **PASS** | **PASS** | **FAIL** | **PASS** | LOW | Fallback |
| RT-BE90U node, wired | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | LOW | Alternate (2.5G) |
| ZenWiFi BT8 node, wired | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | LOW+ | Alternate |
| ZenWiFi BT10 node | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | HIGH | Overspend |
| ZenWiFi BQ16 Pro node | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | HIGH | Overspend |
| ZenWiFi BD5 node | **FAIL** | **PASS** | **PASS** | **PASS** | **PASS** | *n/c* | Reject |
| ZenWiFi BE5000 Pro node | **FAIL** | **PASS** | **PASS** | **PASS** | **PASS** | LOW+ | Reject |
| RT-BE82U node | **FAIL** | **PASS** | **PASS** | **PASS** | **PASS** | LOW | Reject |
| RT-BE58 Go node | **FAIL** | **PASS** | **PASS** | **PASS** | **PASS** | LOW | Reject |
| ZenWiFi BN12 node | **PASS** | **PASS** | **PASS** | **PASS** | **PASS** | HIGH | Defer |

| Option | R1 6 GHz | R2 stack | R3 roam | R4 ≥2.5G | R5 no extra gear | R6 cost | Verdict |
|---|---|---|---|---|---|---|---|
| Third-party tri-band kit as router | **PASS** | **FAIL** | **PASS** | *n/c* | **PASS** | *n/c* | Reject |
| Third-party tri-band kit in AP mode | **PASS** | **PASS** | **PART** | *n/c* | **PASS** | *n/c* | Reject |
| UniFi standalone APs | **PASS** | **PASS** | **FAIL** | **PASS** | **FAIL** | LOW | Reject |
| UniFi managed APs behind the RT-BE92U | **PASS** | **PASS** | **PART** | **PASS** | **FAIL** | MED | Reject |
| UniFi full swap | **PASS** | **FAIL** | **PASS** | **PASS** | **PART** | HIGH | VLAN rebuild only |
| Flint 3 as AP behind the RT-BE92U | **PASS** | **PASS** | **PART** | **PASS** | **PASS** | LOW | Reject |
| Flint 3 replaces the primary | **PASS** | **FAIL** | **PART** | **PASS** | **PASS** | LOW | Cheaper VLAN rebuild |
| Flint 3 pair, GL.iNet mesh | **PASS** | **FAIL** | *n/c* | **PASS** | **PASS** | MED | Reject |
| Flint 4 in any role | **PASS** | *n/c* | *n/c* | **PASS** | **PASS** | HIGH | Needs US approval |

Cost tokens are relative: LOW, up to about one RT-BE92U (MSRP $249.99); LOW+ slightly more; MED adds controller or PoE gear; HIGH multiplies unit cost. R4 for wireless backhaul is inferred from lon.tv's ZenWiFi measurements; the BN12 row rests on its announced AiMesh support; the BE5000 Pro row prices its only listing, a Best Buy 3-pack; Flint rows follow the roles GL.iNet documents (§5.3). The wired RT-BE92U and RT-BE90U rows tie; the 10G port and the matched pair decide for the RT-BE92U (§4). Figure 7 lists the single-unit price snapshots behind most cost tokens and Figure 8 sets price against the 6 GHz ceiling.

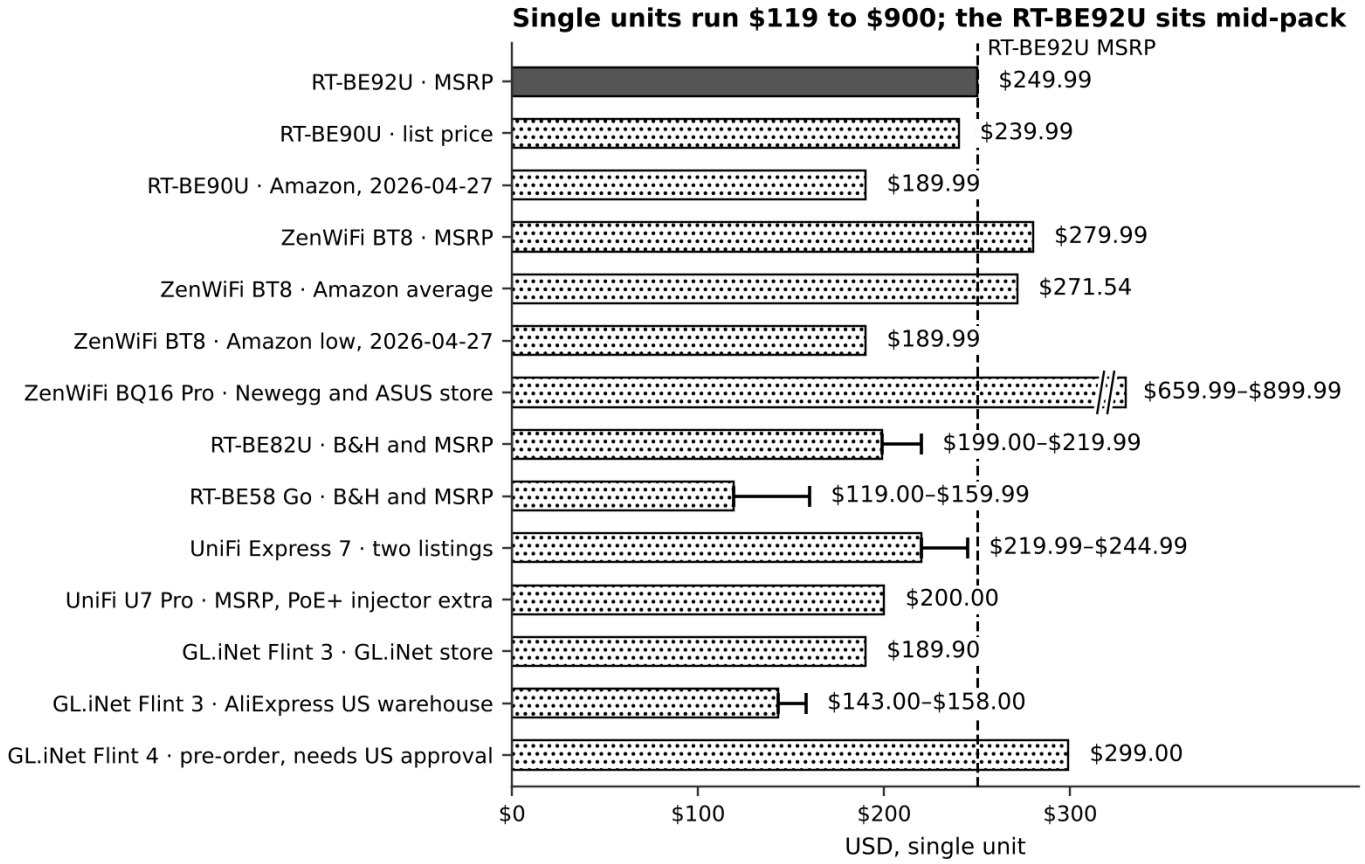



*Figure 7. USD price snapshots for single units: SHI MSRPs, camelcamelcamel's Amazon history, Altex listings, GL.iNet's store and AliExpress listings, and the Flint 4 pre-order; for the RT-BE90U, BQ16 Pro, RT-BE82U and RT-BE58 Go, list prices and the listings of ASUS's US store, Newegg, B&H and Amazon. The BQ16 Pro bar is cut short. Listings in S$ and £ (BT10, BN12) are omitted rather than converted, as is the BE5000 Pro, sold only as a 3-pack.*

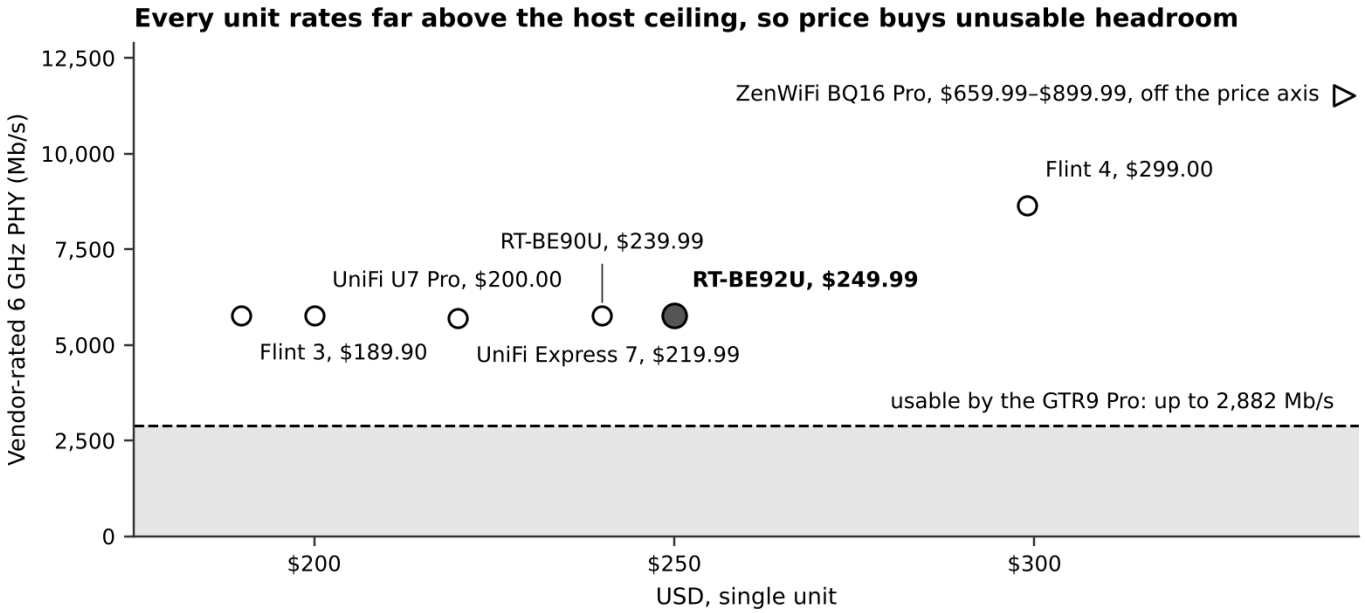



*Figure 8. Single-unit price against the vendor-rated 6 GHz PHY, for the options with a US dollar price. Roles differ: the U7 Pro is a ceiling AP needing PoE+, the Flints are routers (§5). The BQ16 Pro, at $659.99 to $899.99, lies off the price axis (arrow). The ZenWiFi BT8 is omitted because ASUS lists no separate 6 GHz rate for it; the BT10 and BN12 are omitted for want of a US price; the dual-band nodes have no 6 GHz radio.*

# 8 Target architecture

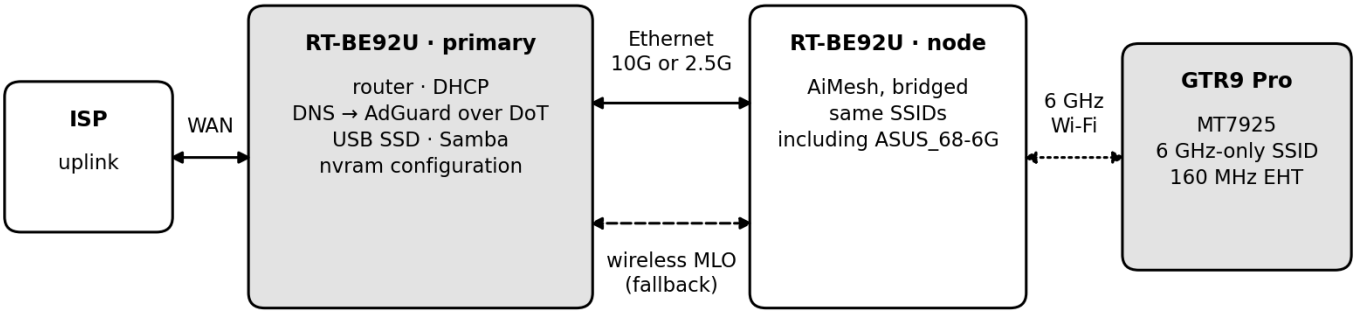



*Figure 9. Target topology. Dashed: wireless MLO backhaul, fallback only.*

The primary keeps routing, DHCP, DNS (AdGuard over DoT) and the Samba share; the node only bridges, and the host keeps plaintext DHCP DNS as designed. AiMesh roams clients across nodes; checks V1 and V4 verify that the node carries ASUS_68-6G (Figure 9).

# 9 Deployment plan

| Step | Action | Done when |
|---|---|---|
| P0 Baseline | Record V1–V3 at the weak spot, V5–V7 anywhere and the primary's firmware version (Appendix A) | Values saved; V1's BSSID test waits for P5 |
| P1 Backhaul path | Confirm an Ethernet or coax route (U1) and the WAN port (U2); pick a panel of Figure 2 | Path and ports chosen |
| P2 Procure | One RT-BE92U (MSRP $249.99); Cat6A for 10G, Cat5e or better for 2.5G | Unit and cable on hand |
| P3 Prepare the node | Update it to the primary's stock release line, then factory-reset it for pairing | Node at defaults, same release line |
| P4 Protect the primary | Save the configuration before any change (nvram save); it lives only in nvram and is lost on factory reset | Backup stored off the router |
| P5 Pair | Add the node from the AiMesh page or the ASUS Router app; cable the chosen primary port to the node's 10G or 2.5G WAN/LAN port | Node listed in AiMesh |
| P6 Pin the backhaul | Set the node's Backhaul Connection Priority (AiMesh → Topology → node → Management) to "Ethernet first", which falls back to wireless; once V8 shows Ethernet and airtime matters more than fallback, switch to Ethernet Backhaul Mode (AiMesh → System Settings) | V8 shows Ethernet backhaul |
| P7 Verify | Re-run V1–V8 at the P0 spots | Every pass criterion met |
| P8 Roll back | Remove the node from AiMesh; restore the P4 backup if a primary setting changed | Primary matches P0 |

# 10 Verification plan

The first table maps each requirement of §3 to the checks that settle it. Commands run on the GTR9 Pro unless noted; values in angle brackets are placeholders, and `iw dev` lists the wireless interface name. None of these commands change host configuration. Record results in Appendix A.

| Requirement (§3) | Field check | When | If it fails |
|---|---|---|---|
| R1 6 GHz coverage on the host's SSID | V1 | P7 | Re-site the node; confirm the SSID reaches it |
| R2 Asuswrt stack untouched | V5, V6, V7 | P0 and P7 | Roll back (P8); the node must not serve DHCP or DNS |
| R3 One roaming domain | V4 | P7 | Check both units are in AiMesh, not bridged |
| R4 Backhaul of 2.5 Gb/s or more | V2, V8 | P7 | Re-seat the cable; re-check the port pair (Figure 2) |
| R5 No controller or extra gear | none; set at purchase | P2 | Not applicable to an AiMesh node |
| R6 Low incremental cost | none; set at purchase | P2 | Re-check prices (U6) |

| ID | Check | Command | Pass criterion |
|---|---|---|---|
| V1 | 6 GHz association via the node | `iw dev <iface> link` | freq between 5,925 and 7,125 MHz; tx bitrate shows "160MHz" and "EHT-MCS" (iw output); after P5 the BSSID near the node differs from the primary's |
| V2 | Throughput, both directions | `# wired LAN host`<br>`iperf3 -s`<br>`# GTR9 Pro`<br>`iperf3 -c <server> -t 30 -P 4`<br>`iperf3 -c <server> -t 30 -P 4 -R` | Both directions at or above the P0 baseline at the same spot |
| V3 | Latency and loss to the primary | `ping -c 100 <primary-lan-ip>` | 0% loss; avg and mdev at or below the baseline |
| V4 | Roaming | `iw event -t` | Walk from the primary to the node and back: a new association to the node's BSSID appears; any disconnect-to-association gap in the timestamps is under 1 s |
| V5 | DNS invariant | `resolvectl status <iface>`<br>`resolvectl query example.com` | DNS server is the primary's LAN IP from DHCP; the link shows -DNSOverTLS; the query resolves; identical to P0 |

| ID | Check | Command | Pass criterion |
|---|---|---|---|
| V6 | USB share invariant | `smbclient -L //<primary-lan-ip> -U <user>` | Share listed, identical to P0 (needs the smbclient package) |
| V7 | DHCP lease invariant | `ip -br addr show dev <iface>` | Address still in the primary's LAN subnet |
| V8 | Backhaul type (router) | Primary web GUI → AiMesh → Topology → node; no CLI equivalent verified | The node reports Ethernet backhaul on a wired build |

# 11 Risks and unknowns

| ID | Item | Impact | Resolution | When |
|---|---|---|---|---|
| U1 | Ethernet or coax path to the node location | Decides wired or wireless backhaul | Survey the route; MoCA if only coax exists | Before buying |
| U2 | Which RT-BE92U port carries WAN; ISP plan rate | Decides 10G or 2.5G backhaul | Port plan (Figure 2) | Before buying |
| U3 | WPA/WPA2-Enterprise in use | AiMesh mode disables it | Confirm it is unused; otherwise an AP-based design | Before buying |
| U4 | Wireless-fallback airtime | Host link and backhaul leg share the node's 6 GHz radio (§6.2) | Prefer wired; quantify with V2 | At install |
| U5 | Firmware | One release line keeps AiMesh in lockstep, but a model-specific regression reaches both units | Record versions at P0 and P7; defer non-security updates briefly; keep the P4 backup | Ongoing |
| U6 | Prices | Listing snapshots | Re-check at purchase | Before buying |
| U7 | BT8 port conflict | ASUS lists dual 2.5G; one reseller lists dual 10G and 18 Gb/s | The ASUS specification governs | Only if BT8 |
| U8 | UniFi storage share | No SMB share feature identified; RTINGS reports no USB port on the UDR7, while one retailer lists USB 3.0 | Matters only for a full swap; confirm on ui.com before buying | Only if UniFi |
| U9 | BN12 and Wi-Fi 8 | No ETA at Scan; Pokde.Net reports an October launch, Dong Knows reports 2027; Wi-Fi 8 gains need Wi-Fi 8 clients, and the final IEEE specification can change features; US sale needs conditional approval (U12) | Revisit when Wi-Fi 8 clients exist and it is on US sale | Later |
| U10 | Other clients | Phone and laptop Wi-Fi not assessed | Node coverage helps them; verdict unchanged | Later |
| U11 | Segmentation without a platform change | Smart Home Master sets up to three SSIDs; isolation not assessed | Test before treating VLAN needs as a reason to switch (§12) | Later |
| U12 | US router rule (FCC Covered List) | Since 2026-03-23, new foreign-made consumer router models cannot be authorized for US import or sale without conditional approval; models approved earlier stay on sale, with update waivers to at least 2027-03-01, reportedly extended to January 2029. GL.iNet says its US-sold models are unaffected; the BN12 and Flint 4 were announced after the cut-off | Buy models already on US sale (RT-BE92U, RT-BE90U, BT8, Flint 3); check the FCC ID of any model announced after March 2026 | Before buying |
| U13 | GL.iNet mesh and roaming | "Mesh" is listed for the Flint 3; roaming method and partner models undocumented | Verify before any Flint pair | Only if Flint |
| U14 | Flint DoT to a private AdGuard DNS endpoint | Custom DoT or DoH servers need SSH and are not officially supported | Treat as rebuild cost; test on firmware 4.8.x | Only if Flint |
| U15 | Mixed-model AiMesh (RT-BE90U, BQ16 Pro) | Listed as AiMesh nodes, with no published test under an RT-BE92U; MLO backhaul and the BQ16 Pro's second 6 GHz radio are documented only between same-model units | Prefer wired backhaul; confirm V1 and V8 at P7 | Only if either |

| ID | Item | Impact | Resolution | When |
|---|---|---|---|---|
| U16 | RT-BE82U band conflict | An ASUS overview graphic shows a 6 GHz rate; the specification, FCC grant and reviews list 2.4 and 5 GHz only | The ASUS specification governs | Only if RT-BE82U |

## 12 What would change the verdict

- **A cable reaches the GTR9 Pro's location:** wire the host directly; the node then serves the other clients.
- **The backhaul will be 2.5G either way** (WAN keeps the 10G port, or the run cannot carry 10G): the RT-BE90U meets every requirement for $10 less, without the matched pair (U15).
- **Segmentation becomes a requirement:** test Smart Home Master SSIDs first (U11); if true VLANs are needed, the cheaper rebuild is a Flint 3 (VLANs through LuCI or UCI; DoT, Samba and ad blocking built in, §5.3) and the managed one is UniFi (§5.2); both discard the R2 stack.
- **Enterprise Wi-Fi authentication becomes necessary:** AiMesh is out; move to an AP-based design.
- **The node must sit in a living space:** ZenWiFi BT8 instead, accepting its 2.5G wired ceiling.
- **Wi-Fi 8 clients arrive and the BN12 is on US sale:** re-evaluate with a Wi-Fi 8 primary; Multi-AP Coordination ties router and nodes, so a Wi-Fi 7 primary would gate it (inference).

## 13 Method and abbreviations

Facts about this network are as stated and were not re-measured. Vendor figures are as listed by manufacturers and retailers; reviews, listings and regulatory notices were checked from 2026-10-01 to 2026-10-03, and on 2026-10-08 for the five models added in v1.7; this edition does not reproduce the source list. The host PHY rate is computed as 1,960 data tones × 12 bits × 5/6 ÷ 13.6 µs × 2 spatial streams = 2,882 Mb/s (EHT, 160 MHz, MCS 13, 0.8 µs GI); other PHY rates are vendor ratings. Prices are listing snapshots, not checkout quotes. Inferences are marked as such.

### Abbreviations

| Term | Meaning | Term | Meaning |
|---|---|---|---|
| **AP** | access point | **BSSID** | MAC identifier of one access point radio |
| **DHCP** | protocol that hands clients an address and DNS server | **DNSSEC** | signature validation for DNS answers |
| **DoH** | DNS over HTTPS | **DoT** | DNS over TLS |
| **EHT** | Extremely High Throughput (802.11be, Wi-Fi 7) | **GI** | guard interval between symbols |
| **FCC** | US Federal Communications Commission | **ISP** | internet service provider |
| **LuCI, UCI** | OpenWrt web interface and command-line configuration | **MCS** | modulation and coding scheme |
| **MLO** | Multi-Link Operation: one Wi-Fi 7 link across bands | **MoCA** | Multimedia over Coax Alliance networking |
| **MSRP** | manufacturer's suggested retail price | **nvram** | router's non-volatile settings store |
| **PHY** | physical layer; PHY rate is the raw link rate | **PoE, PoE+** | Power over Ethernet (802.3af, 802.3at) |
| **SFP+** | pluggable 10G transceiver port | **SMB** | Windows file-sharing protocol (Samba) |
| **SSH** | secure shell command-line access | **SSID** | Wi-Fi network name |
| **VLAN** | virtual LAN: a separated segment over shared wiring | **WDS** | Wireless Distribution System bridging |

**This edition.** v1.7 (2026-10-08): the ASUS RT-BE90U, ZenWiFi BQ16 Pro, ZenWiFi BE5000 Pro, RT-BE82U and RT-BE58 Go assessed as AiMesh nodes; the decision matrix and Figure 6 now rank twenty options; §1, §4, §6.2, §7, §12, §13, Figures 5, 7 and 8, U12 and the new risks U15 and U16 cover the additions. Earlier editions are listed in the repository CHANGELOG.

# A Field worksheet

Print this page. Record each check at P0, before buying, and again at P7, after pairing; pass criteria are in §10. V4 and V8 have no P0 value: both need the node in place.

| | | | |
|---|---|---|---|
| **Date** | | **Node location** | |
| **Primary firmware** | | **Node firmware** | |
| **ISP plan** | | **Backhaul** | ☐ 10G ☐ 2.5G ☐ MoCA ☐ wireless |

| ID | Check | Record | P0 baseline | P7 after | Pass |
|---|---|---|---|---|---|
| V1 | 6 GHz link via the node | freq; tx bitrate; BSSID | | | ☐ |
| V2 | Throughput | Mb/s down / up | | | ☐ |
| V3 | Latency and loss | loss %; avg / mdev ms | | | ☐ |
| V4 | Roaming | longest gap (s) | *n/a* | | ☐ |
| V5 | DNS invariant | DNS server; DoT flag | | | ☐ |
| V6 | USB share | share listed | | | ☐ |
| V7 | DHCP lease | address and subnet | | | ☐ |
| V8 | Backhaul type | Ethernet or wireless | *n/a* | | ☐ |

| Notes: channel or width changes, BSSID flaps, firmware versions, dead spots |
|---|
| |
| |
| |
| |

Outcome ☐ keep the node ☐ re-site it and re-run V1–V3 ☐ roll back (P8)